

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,B,D	B,C	A,C	A,B,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	A	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.33	55.55	8.88	53.75	1.30	1.44	0.16	0.48

PART-2 : CHEMISTRY

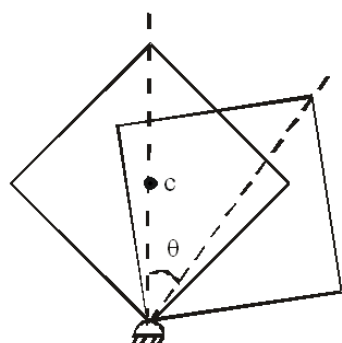
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	A,B,D	A,B,D	B,C,D	A,B,D	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.69 or 0.70	23.40	10.00	20.00	5.00	13.66 to 13.67	247.74	16.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,C	A,C,D	A,B,C	A,B,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	18.00	1.00	3.00	240.00	0.75	3.00	4.00	9.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,C)



$$Mg \frac{a}{\sqrt{2}} (1 - \cos \theta) = \frac{1}{2} \left(\frac{Ma^2}{6} + \frac{Ma^2}{2} \right) \omega^2$$

$$\omega^2 = \frac{3g}{a\sqrt{2}} (1 - \cos \theta)$$

$$mg \cos \theta = m\omega^2 \frac{a}{\sqrt{2}} \text{ (as no reaction towards hinge)}$$

$$\cos \theta = \frac{3}{5} \Rightarrow \theta = 53^\circ$$

when 'c' is at same level as hinge

$$mg \frac{a}{\sqrt{2}} = \left(\frac{ma^2}{6} + \frac{ma^2}{2} \right) \alpha \Rightarrow \alpha = \frac{3g}{2\sqrt{2}a}$$

$$a_{cm} = \frac{3g}{2\sqrt{2}a} \cdot \frac{a}{\sqrt{2}} \Rightarrow a_{cm} = \frac{3g}{4}$$

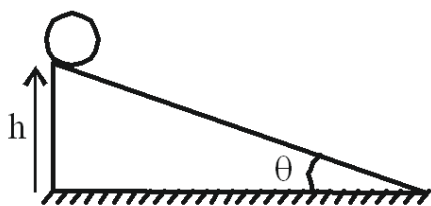
$$mg - R_y = ma_m \Rightarrow R_y = \frac{mg}{4}$$

$$\text{also } R_x = m\omega^2 \frac{a}{\sqrt{2}} \text{ net rxn} = \sqrt{R_x^2 + R_y^2}$$

 for max ω

$$mg a\sqrt{2} = \frac{1}{2} \left(\frac{2}{3} ma^2 \right) \omega$$

2. Ans (A,B,D)



$$(\mu mg \cos \theta)R = \frac{2}{5}mR^2 \frac{a}{R}$$

$$a = \frac{5g \sin \theta}{7} \quad (\text{for pure rolling})$$

$$(\mu mg \cos \theta)R = \frac{2}{5}mR^2 \frac{5g \sin \theta}{7R}$$

$$\mu = \frac{2}{7} \tan \theta$$

$$\text{If } \mu = \frac{\tan \theta}{7}$$

$$a_{\text{cm}} = g \sin \theta - \mu g \cos \theta$$

$$a_{\text{cm}} = \frac{6g \sin \theta}{7}$$

$$\therefore \frac{h}{\sin \theta} = \frac{1}{2} \frac{6g \sin \theta}{7} t^2$$

$$t^2 = \frac{7h}{3g \sin^2 \theta} \Rightarrow t = \frac{1}{\sin \theta} \sqrt{\frac{7h}{3g}}$$

$$R(\mu mg \cos \theta) = \frac{2}{5}mR^2 \alpha$$

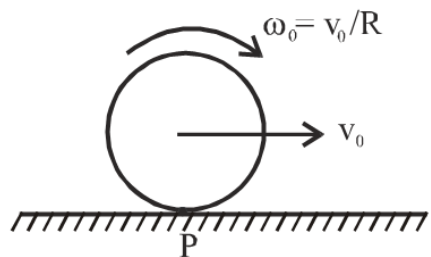
$$R \left(\frac{\mu mg \cos \theta}{7} \right) = \frac{2}{5}mR^2 \alpha$$

$$\alpha = \frac{5g \sin \theta}{14R}$$

3. Ans (B,C)

as the direction of angular speed is not given
hence

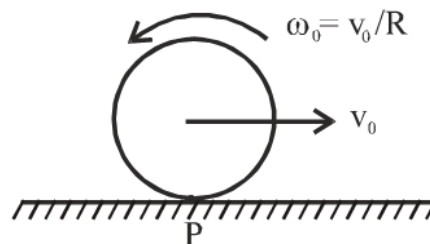
Case (1)



$$L_P = mv_0 R + \frac{mR^2}{2} \frac{v_0}{R}$$

$$= \frac{3}{2}mv_0 R$$

Case (2)



$$L_P = mv_0 R - \frac{mR^2}{2} \frac{v_0}{R}$$

$$= \frac{mv_0 R}{2}$$

4. Ans (A,C)

$$\mu = \frac{c_0}{c} = \frac{3}{2} = 1.5$$

$$a = 10 \text{ cm}, t = 2 \text{ mm}$$

$$R^2 = (R - t)^2 + \frac{a^2}{4}$$

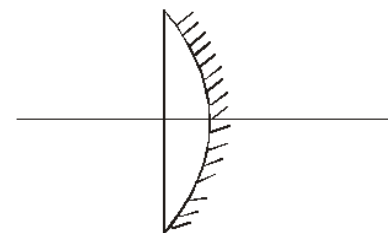
$$R = \frac{500}{8} = \frac{125}{2} \text{ cm}$$

5. Ans (A,B,D)



$$\frac{1}{30} = 2 \left(\frac{3}{2} - 1 \right) \left[\frac{1}{R} - \frac{1}{\infty} \right] - \frac{2}{\infty}$$

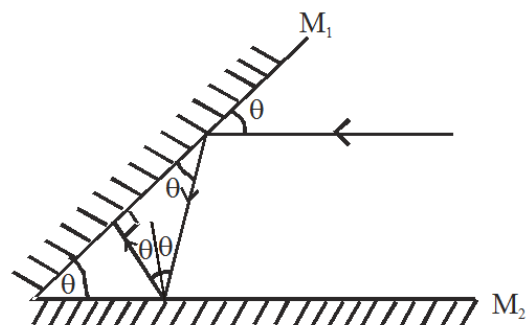
$$R = 30 \text{ cm}$$



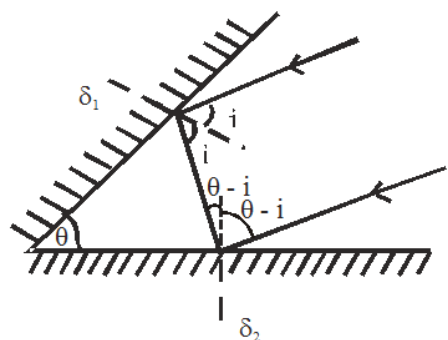
$$\frac{1}{F} = 2 \left(\frac{3}{2} - 1 \right) \left[\frac{1}{\infty} - \frac{1}{-30} \right] - \frac{2}{-30}$$

$$\frac{1}{F} = \frac{1}{30} + \frac{1}{15} \Rightarrow F = 10 \text{ cm}$$

6. Ans (A,C)



$$3\theta = 90^\circ \Rightarrow \theta = 30^\circ$$



$$\delta_1 = 180 - 2i$$

$$\delta_2 = 180 - 2(\theta - i)$$

$$\therefore \delta_{\text{net}} = \delta_1 + \delta_2 = 360 - 2\theta$$

Does not depend on i .

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)

By angular momentum conservation about the hinge,

$$L_i = L_f$$

$$\therefore mv \frac{L}{2} = \frac{mL^2}{3} \omega$$

$$\therefore \omega = \frac{3v}{2L} \quad ; \quad t = \frac{2L}{3v}$$

$$v_{\text{CM}} = \frac{\omega L}{2} = \frac{3v}{4}$$

$$KE_f = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times \frac{mL^2}{3} \times \left(\frac{3v}{2L} \right)^2$$

$$= \frac{3}{8} mv^2$$

$$\text{Impulse} = m \left(v - \frac{3v}{4} \right) = \frac{mv}{4}$$

$$\therefore F_{\text{hinge}} = \frac{mv}{4} \div \frac{L}{100v}$$

$$= 25 \frac{mv^2}{L}$$

$$\text{Later, } F_{\text{hinge}} = m\omega^2 \frac{L}{2}$$

$$= m \left(\frac{3v}{2L} \right)^2 \times \frac{L}{2}$$

$$= \frac{9mv^2}{8L}$$

8. Ans (A)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\therefore \frac{1}{v} + \frac{1}{-3f} = \frac{1}{f}$$

$$\therefore v = \frac{3f}{4}$$

$$\frac{dv}{dt} = -\frac{v^2}{u^2} \times \frac{du}{dt}$$

$$= -\left(\frac{1}{4} \right) \times 1$$

$$= -\frac{1}{16} \text{ m/s}$$

$$y_{\text{img}} = y_{\text{obj}} \times m$$

$$= y_{\text{obj}} \times \left(1 - \frac{v}{f} \right)$$

$$\therefore \frac{dy_{\text{img}}}{dt} = \frac{dy_{\text{obj}}}{dt} \left(1 - \frac{v}{f} \right) - \frac{y_{\text{obj}}}{f} \times \frac{dv}{dt}$$

$$= 1 \left(1 - \frac{3}{4} \right) - 0$$

$$= \frac{1}{4} \text{ m/s.} = -4 \times \frac{dv}{dt}$$

$\therefore$ The image moves along $4x + y = 3f$

9. Ans (B)

By angular momentum conservation $L_1 = L_f$

about impact point.

$$\therefore mu \times \frac{5R}{6} + \frac{mR^2}{2} \times \frac{u}{R} = \frac{3}{2}mR^2\omega_0$$

$$\therefore \omega_0 \frac{8u}{9R}$$

$$= \frac{10}{9} \text{ rad/s if } u = 1 \text{ m/s}$$

Using energy conservation for the rotation

about the point of impact :

$$\frac{1}{2} \times \frac{3}{2}mR^2 \times \omega_0^2 = mg \times \frac{R}{6}$$

$$\therefore \frac{3}{4}mR^2 \left(\frac{8u_1}{9R} \right)^2 = \frac{mgR}{6}$$

$$\therefore u_1 = \sqrt{\frac{9gR}{32}} = 1.5 \text{ m/s}$$

By Newton's Law

$$mg \cos \theta - N = m\omega_0^2 R$$

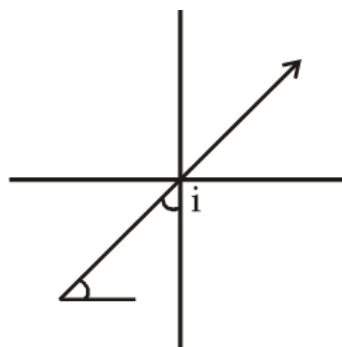
$$\therefore N = mg \times \frac{5}{6} - m \left(\frac{8u_2}{9R} \right)^2 \times R$$

$$= 0 \text{ if } u_2^2 = \frac{9gR}{32} \times \frac{15}{4}$$

$$\therefore u_2 = 1.5 \times \frac{3.88}{2} = 2.91$$

$$J_x = m\omega_0 R \cos \theta - mu = -\frac{7}{24}Ns$$

10. Ans (D)



By shell's Law :

$$\mu \sin i = \text{constant}$$

$$= 1 \text{ here}$$

$$\therefore \mu = \sec \theta$$

$$= \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

Case 1 : $\frac{dy}{dx}$ changes from 0 to ∞ as y changes

from 0 to R .

Hence the circle must be $x^2 + (y - R)^2 = R^2$

$$\frac{dy}{dx} = \frac{x}{R - y}$$

$$\therefore \mu = \frac{R}{R - y} = \frac{4}{3} \text{ at } y = \frac{R}{4}$$

$$\text{If } y = \frac{2R}{5}, x^2 + \left(\frac{2R}{5} - R \right)^2 = R^2 \Rightarrow x = \frac{4R}{5}$$

Case 2 : The circle must be $(x - R)^2 + y^2 = R^2$

$$\text{If } y = \frac{3R}{5}, (x - R)^2 + \left(\frac{3R}{5} \right)^2 = R^2$$

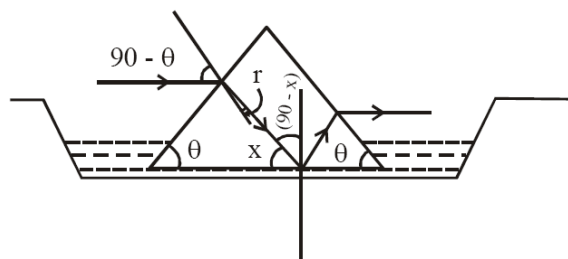
$$\therefore x = \frac{R}{5}$$

$$\text{Here, } \frac{dy}{dx} = \frac{R - x}{y}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.33)



$$\sin(90 - \theta) = \mu_g \sin r$$

$$\cos \theta = \frac{5}{3} \sin r \quad \dots(1)$$

$$\sin r = \frac{3}{5} \cos \theta \quad \dots(2)$$

for TIR

$$(90 - x) > C$$

$$(r + \theta) > C$$

two inside angle equal to exterior angle

$$\sin(r + \theta) > \sin C$$

$$\sin r \cos \theta + \cos r \sin \theta > \frac{4}{5}$$

$$\left[\sin C = \frac{1}{\mu_g} = \frac{1}{\mu_w} = \frac{1}{\frac{5}{3}} \times \frac{4}{3} = \frac{4}{5} \right]$$

From equation (2)

$$\frac{3}{5} \cos^2 \theta + \frac{\sqrt{25 - 9\cos^2 \theta}}{5} \sin \theta > \frac{4}{5}$$

$$\sqrt{25 - 9\cos^2 \theta} \sin \theta > 4 - 3\cos^2 \theta$$

$$(25 - 9\cos^2 \theta) (1 - \cos^2 \theta) > (4 - 3\cos^2 \theta)^2$$

$$(25 - 25\cos^2 \theta - 9\cos^2 \theta + 9\cos^4 \theta) > \{16 + 9\cos^4 \theta - 24\cos^2 \theta\}$$

$$25 - 34\cos^2 \theta > 16 - 24\cos^2 \theta$$

$$9 > 10\cos^2 \theta$$

$$\frac{9}{10} > \cos^2 \theta$$

$$\frac{10}{9} < \sec^2 \theta$$

$$\frac{10}{9} < 1 + \tan^2 \theta$$

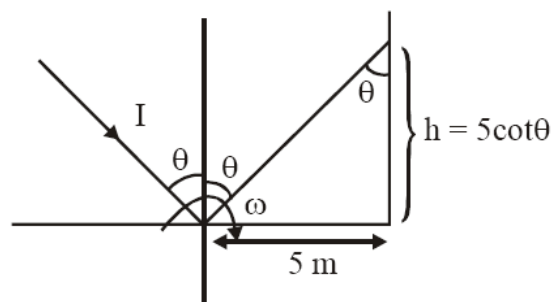
$$\tan^2 \theta > \frac{1}{9}$$

$$\tan \theta > \frac{1}{3}$$

$$\theta > \tan^{-1} \left(\frac{1}{3} \right)$$

$$x = 0.33$$

2. Ans (55.55)



When mirror is rotated with angular speed ω the reflected ray rotates with angular speed ' 2ω '

$$\text{speed of spot} = \frac{dh}{dt}$$

$$= \frac{d}{dt} (5 \cot \theta)$$

$$= \left| -5 \csc^2 \theta \frac{d\theta}{dt} \right|$$

$$= \left| -5 \csc^2 37^\circ \times 4 \right|$$

$$= \frac{5}{(0.6)^2} \times 4$$

$$= 55.55 \text{ m/s}$$

3. Ans (8.88)

$$\frac{\text{Mass of disc c}}{\text{Mass of full disc}} = \frac{\pi \left(\frac{R}{3} \right)^2}{\pi R^2} = \frac{1}{9}$$

$$\text{Mass of disc 'C'} = \frac{M}{9}$$

$$\text{Mass of K} = M - \frac{M}{9} = \frac{8M}{9}$$

$$OO_2 \times \frac{8M}{9} = \frac{2R}{3} \times \frac{M}{9}$$

$$OO_2 = \frac{R}{12}$$

$$I_{CO_2} = \frac{1}{2} \left(\frac{M}{9} \right) \left(\frac{R}{3} \right)^2 + \left(\frac{M}{9} \right) \left(\frac{2R}{3} + \frac{R}{12} \right)^2$$

$$= \frac{MR^2}{162} + \frac{M}{9} \times \frac{81R^2}{144}$$

$$= \frac{MR^2}{9} \left(\frac{1}{18} + \frac{81}{144} \right) = \frac{MR^2}{9} \times \frac{89}{144}$$

Moment of inertia of full disc about O_2

$$= \frac{1}{2} MR^2 + M \left(\frac{R}{12} \right)^2,$$

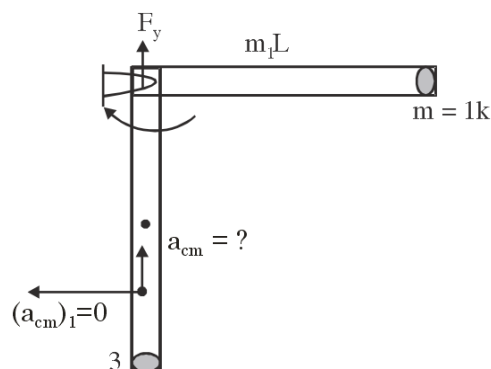
$$= \frac{1}{2} MR^2 + \frac{MR^2}{144} = \frac{73}{144} MR^2$$

So MOI of disc K about O_2 is

$$I_K = I_{\text{full disc}} - I_C$$

$$\begin{aligned}
 &= \frac{73}{144}MR^2 - \frac{89MR^2}{144 \times 9} \\
 &= \frac{MR^2}{144} \left\{ 73 - \frac{89}{9} \right\} \\
 &= \frac{568MR^2}{144 \times 9} = \frac{71}{162}MR^2 \\
 &= \frac{71}{162} \times 81 \times \left(\frac{1}{2} \right)^2 \\
 &= \frac{71}{2} \times \frac{1}{4} \\
 &= 8.875 \simeq 8.88
 \end{aligned}$$

4. Ans (53.75)



$$\Delta K + \Delta u = 0$$

$$\frac{1}{2}I\omega^2 + \left(-MgL - Mg\frac{L}{2} \right) = 0$$

$$\frac{1}{2} \left(\frac{ML^2}{3} + ML^2 \right) \omega = \frac{3}{2}MgL$$

$$\frac{1}{2} \left(\frac{4ML^2}{3} \right) \omega = \frac{3}{2}MgL$$

$$\omega = \frac{3}{2} \sqrt{\frac{g}{L}}$$

Centre of Mass lie at $\frac{3L}{4}$

$$C_{\text{Hinge}} = O \Rightarrow \alpha = 0$$

$$(a_{\text{cm}})_c = \omega^2 \left(\frac{3L}{4} \right)$$

$$= \left(\frac{9}{4} \frac{g}{L} \right) \left(\frac{3L}{4} \right)$$

$$= \frac{27g}{16}$$

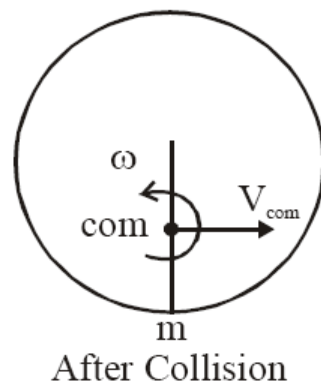
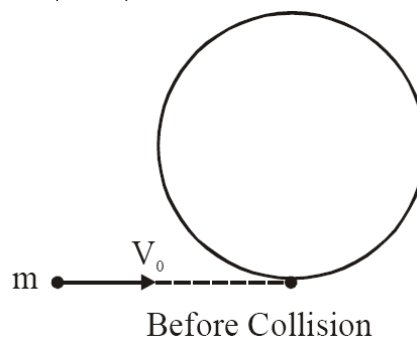
$$F_y - 2mg = 2ma_c$$

$$F_y = 2mg + 2m \left(\frac{27}{16}g \right)$$

$$= 2mg + \frac{27mg}{8}$$

$$F_y = \frac{46mg}{8} = \frac{43}{8} \times 10 = \boxed{53.75\text{N}}$$

5. Ans (1.30)



Momentum Remain conserved

$$mv_0 = (m + m) v_{\text{cm}}$$

$$v_{\text{cm}} = \frac{v_0}{2}$$

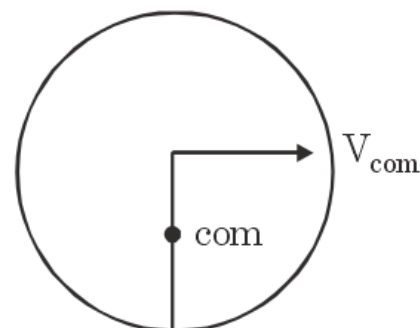
Conservation of angular momentum about a point very near to COM and in air

$$mv_0 \frac{R}{2} = (I_{\text{cm}} \omega) + r \times 2mv_c$$

$$mv_0 \frac{R}{2} = \left\{ \frac{MR^2}{2} + m \left(\frac{R}{2} \right)^2 + m \left(\frac{R}{2} \right)^2 \right\} \omega$$

$$mv_0 \frac{R}{2} = mR^2 \omega$$

$$\omega = \frac{v_0}{2R}$$



$$v_c = v_{\text{cm}} - \omega R$$

$$= \frac{v_0}{2} - \frac{R}{2} \times \frac{v_0}{2R}$$

$$v_c = \frac{v_0}{4} = \frac{5.2}{4}$$

$$= 1.30 \text{ m/s}$$

6. Ans (1.44)

$$mgh = \frac{1}{2} I_{AOR} \omega^2$$

$$mg \frac{\ell}{2} (1 - \sin \theta) = \frac{1}{2} \left(\frac{m\ell^2}{12} + \frac{m\ell^2}{4} \cos^2 \theta \right) \omega^2$$

$$10 \times \frac{2}{2} (1 - \sin 53^\circ) = \frac{1}{2} \left(\frac{4}{12} + \frac{4}{4} \cos^2 53^\circ \right) \omega^2$$

$$10 \left(1 - \frac{4}{5} \right) = \frac{1}{2} \left(\frac{4}{12} + \frac{4}{4} \times \frac{9}{25} \right) \omega^2$$

$$10 \times \frac{1}{5} = \frac{1}{2} \times \frac{4}{12} \left(1 + \frac{27}{25} \right) \omega^2$$

$$2 = \frac{1}{6} \left(\frac{52}{25} \right) \omega^2$$

$$\omega = \sqrt{\frac{75}{13}}$$

$$v_c = \left(\frac{\ell}{2} \cos \theta \right) \omega$$

$$= \left(\frac{\ell}{2} \times \cos \theta \right) \omega$$

$$= \frac{2}{2} \times \cos 53^\circ \times \sqrt{\frac{75}{13}}$$

$$= \frac{3}{5} \times \sqrt{\frac{75}{13}}$$

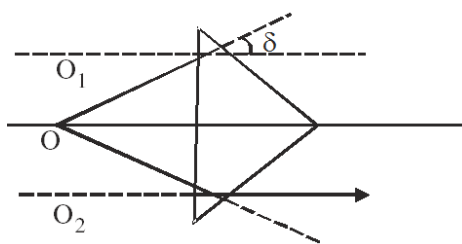
$$= 1.44 \text{ m/s}$$

7. Ans (0.16)

For thin prism $\delta = (\mu - 1) \alpha$

$$= (1.5 - 1) \times 1.8^\circ$$

$$= 0.5 \times 1.8^\circ \times \frac{\pi}{180} \text{ rad} = \frac{\pi}{200} \text{ Rad}$$



$$OO_1 = 10 \times \frac{\pi}{200} = \frac{\pi}{20} \text{ (m = OO}_2\text{)}$$

For Mirror O_1 & O_2 are two point object

$$\frac{1}{V} + \frac{1}{-30} = \frac{1}{-10}$$

$$V = -15 \text{ cm}$$

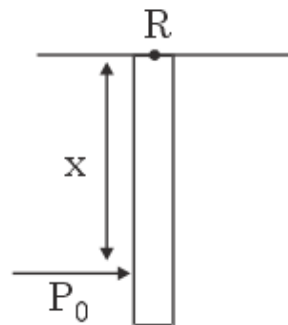
$$m = -\frac{v}{u} = -\frac{(-15)}{(-30)} = -\frac{1}{2}$$

Separation between images

$$= \frac{1}{2} O_1 O_2$$

$$= \frac{1}{2} \times \frac{\pi}{10} = \frac{\pi}{20} \text{ cm} = 0.16 \text{ cm}$$

8. Ans (0.48)



For translational Motion

$$P_0 = M V_{cm}$$

For Rotational Motion

$$P_0 X = I_R \omega$$

$$\omega = \frac{P_0 X}{\frac{ML^2}{3}}$$

$$\text{Initial speed of COM of Rod is } = \omega \times \frac{L}{2}$$

$$= \frac{3P_0 X}{ML^2} \times \frac{L}{2} = \frac{3P_0 X}{2ML}$$

Impulse delivered by pivot is

$$I = m v_{cm} - m \omega \frac{L}{2}$$

But impulse is zero

$$m v_{cm} - m \omega \frac{L}{2} = 0$$

$$V_{cm} = \omega \frac{L}{2}$$

$$\frac{P_0}{M} = \frac{3P_0 X}{2ML}$$

$$X = \frac{2L}{3}$$

$$X = \frac{2}{3} \times 0.72$$

$$X = 0.48 \text{ m}$$

PART-2 : CHEMISTRY

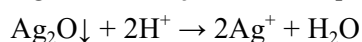
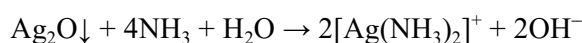
SECTION-I (i)

3. Ans (A,B,D)

SnCl_4 does not give chromyl chloride test

4. Ans (B,C,D)

Fe^{2+} responds to this test but not Fe^{3+} ; Fe(II) gives soluble red iron(II) dimethylglyoxime in alkaline solution



5. **Ans (A,B,D)**

$$60R = i \times \frac{16.2}{162} \times R \times 300$$

$i =$ (AB type solid)

$$z = \frac{d \times a^3 \times N_A}{M}$$

$$= \frac{40 \times 3 \times 3 \times 3 \times 10^{-24} \times 6 \times 10^{23}}{162}$$

$= 4$ (4AB per unit cell)

(O.V. at body centre is empty)

6. **Ans (A,B,D)**

$$\bullet r_{\text{eff}} = \frac{P \cdot N_A \cdot A}{\sqrt{2\pi MRT}}$$

$$\bullet \frac{r_{\text{He}}}{r_{\text{Co}}} = 2.65 \quad \bullet \frac{r_{\text{He}}}{r_{\text{CO}_2}} = 3.31$$

$$\bullet \frac{r_{\text{He}}}{r_{\text{SO}_2}} = 4$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A)**

$$A_v \cdot KE \propto n$$

$$\propto T$$

$$\text{collision frequency} \propto \text{average velocity} \\ \propto n/v$$

8. **Ans (C)**

$$\text{Pressure correction} = \frac{a \cdot n^2}{V^2} = \frac{4 \times 5^2}{10^2} = 1 \text{ atm}$$

$$\text{Ideal volume} = V - nb = 10 - 5 \times 0.05 = 9.75 \text{ L}$$

$$\text{Volume occupied by molecules} = nb/4 = 0.25/4 \\ = 0.0625 \text{ L}$$

$$\text{Volume correction} = nb = 0.25 \text{ L}$$

9. **Ans (B)**

Colour of the bead

Ion

1. Blue green

Cu^{2+}

2. Yellow

Fe^{3+}

3. Green

Cr^{3+}

4. Violet

Mn^{2+}

10. **Ans (A)**

(A) $P \rightarrow 2, Q \rightarrow 1, R \rightarrow 4, S \rightarrow 3$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (0.69 or 0.70)**

$$PF = \frac{4 \times \frac{4}{3} \pi r_{A^+}^3 + 4 \times \frac{4}{3} \pi r_{B^-}^3}{[2(r_{A^+} + r_{B^-})]^3}$$

$$= \frac{\frac{16\pi}{3} \left[\left(\frac{r_{A^+}}{r_{B^-}} \right)^3 + 1 \right]}{[2 \left\{ \left(\frac{r_{A^+}}{r_{B^-}} \right) + 1 \right\}]^3}$$

$$= \frac{16\pi \times [(0.5)^3 + 1]}{3 \times [2 \times (0.5 + 1)]^3}$$

$$= 0.698$$

2. **Ans (23.40)**

$$n_{Ar} = \frac{260}{40} = 6.5$$

$$0.750 = \frac{n_{CH_4}}{n_{CH_4} + 6.5}$$

$$n_{CH_4} = \frac{4.875}{0.25}$$

$$= 19.5$$

$$(KE)_{\text{Total}} = 26 \times 3/2 \times 2 \times 300$$

$$= 23.40 \text{ KCal}$$

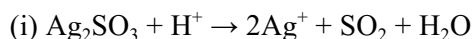
3. **Ans (10.00)**

1, 2, 4, 5, 6, 8, 9, 10, 11, 12

5. **Ans (5.00)**

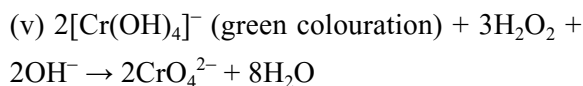
i, ii, v, viii, x

6. Ans (13.66 to 13.67)

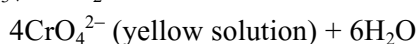
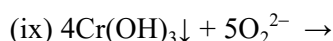
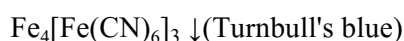
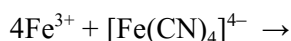
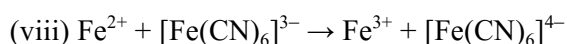


(ii) FeSO_4 solution absorbs NO and becomes brown because of the formation of $[\text{Fe}(\text{NO})]^{2+}\text{SO}_4^{2-}$. (iii) Being covalent do not dissociate

(iv) Ammonium hydroxide in presence of ammonium chloride is group reagent of IIIrd group.

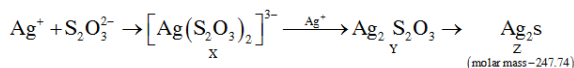


(vi) Red colour solution or precipitate is produced when reagent reacts in alkaline solution.



$$a = 5 \quad b = 4 \quad \frac{a^2 + b^2}{3} = \frac{25 + 16}{3} = 13.67$$

7. Ans (247.74)



8. Ans (16.00)

a is number of black colour sulphide

b is the number of white colour sulphide

c is the number of yellow colour sulphide

PbS , Bi_2S_3 , CuS , HgS , Ag_2S , CoS , NiS – Black (a)

ZnS – White (b)

SnS_2 , As_2S_3 , CdS – Yellow (c)

$$\frac{a^2 - b}{c} = \frac{49 - 1}{3} = 16.00$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C,D)

We have $\vec{B} \times \vec{C} = x\vec{A} + y\vec{B} + z\vec{C} \dots (1)$

Taking dot with $\vec{B} \times \vec{C}$ gives

$$(\vec{B} \times \vec{C}) \cdot (\vec{B} \times \vec{C}) = x[\vec{A} \cdot (\vec{B} \times \vec{C})]$$

$$\Rightarrow x = \frac{(\vec{B} \times \vec{C})^2}{[\vec{A} \cdot (\vec{B} \times \vec{C})]} = \frac{\vec{B}^2 \vec{C}^2 - (\vec{B} \cdot \vec{C})^2}{[\vec{A} \cdot (\vec{B} \times \vec{C})]}$$

Similarly taking dot with $\vec{C} \times \vec{A}$ gives

$$y = \frac{(\vec{B} \times \vec{C}) \cdot (\vec{C} \times \vec{A})}{[\vec{A} \cdot (\vec{B} \times \vec{C})]}$$

and taking dot with $\vec{A} \times \vec{B}$ gives

$$z = \frac{(\vec{B} \times \vec{C}) \cdot (\vec{A} \times \vec{B})}{[\vec{A} \cdot (\vec{B} \times \vec{C})]}$$

$$\text{Now } [\vec{A} \cdot (\vec{B} \times \vec{C})] = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= 1(1 + 1) - 2(-2 - 3) = 12$$

We have $\vec{B}^2 \vec{C}^2 - (\vec{B} \cdot \vec{C})^2 = (6)(2) - (0) = 12$

$$\therefore x = \frac{12}{12} = 1$$

$$y = \frac{\begin{vmatrix} \vec{B} \cdot \vec{C} & \vec{B} \cdot \vec{A} \\ \vec{C} \cdot \vec{C} & \vec{C} \cdot \vec{A} \end{vmatrix}}{12} = \frac{\begin{vmatrix} 0 & -3 \\ 2 & 1 \end{vmatrix}}{12}$$

$$= \frac{0 - (-6)}{12} = \frac{1}{2} \text{ and}$$

$$z = \frac{\begin{vmatrix} \vec{B} \cdot \vec{A} & \vec{B} \cdot \vec{B} \\ \vec{C} \cdot \vec{A} & \vec{C} \cdot \vec{B} \end{vmatrix}}{12} = \frac{\begin{vmatrix} -3 & 6 \\ +1 & 0 \end{vmatrix}}{12} = \frac{0 - (6)}{12}$$

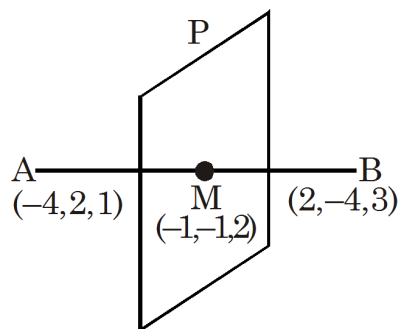
$$= -\frac{1}{2}$$

2. Ans (A,C)

$$\lim_{n \rightarrow \infty} (\sin 1)(1 + \cot^n 1)^{\frac{1}{n}} = \sin 1$$

Now $\sin \frac{\pi}{4} < \sin 1 < \sin \frac{\pi}{3}$

3. Ans (A,C,D)



(A) Plane P will contain the point $M(-1, -1, 2)$ and dr's of its normal will be proportional to $(3, -3, 1)$

$\therefore$ equation of the plane P is

$$3(x+1) - 3(y+1) + 1(z-2) = 0$$

$$P : 3x - 3y + z - 2 = 0 \quad \dots(1)$$

$$Q : x - y + cz - 1 = 0 \quad \dots(2)$$

If P is parallel to Q then

$$\frac{3}{1} = \frac{1}{c} \Rightarrow c = \frac{1}{3}$$

(C) & (D) $\vec{n}_1 = 3\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{n}_2 = \hat{i} - \hat{j} + c\hat{k}$

now

$$\cos 45^\circ = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right| = \left| \frac{3 + 3 + c}{\sqrt{19} \sqrt{2 + c^2}} \right|$$

$$\therefore 2(6 + c^2) = 19(2 + c^2)$$

$$2(36 + c^2 + 12c) = 38 + 19c^2$$

$$17c^2 - 24c - 34 = 0$$

$$\text{product} = -2$$

4. Ans (A,B,C)

$$\lim_{x \rightarrow 0^+} \frac{ae^{1/x} + be^{-1/x}}{ce^{1/x} + de^{-1/x}} = \lim_{x \rightarrow 0^+} \frac{a + be^{-2/x}}{c + de^{-2/x}} = \frac{a}{c}$$

$$\lim_{x \rightarrow 0^-} \frac{ae^{1/x} + be^{-1/x}}{ce^{1/x} + de^{-1/x}} = \lim_{x \rightarrow 0^+} \frac{ae^{2/x} + b}{ce^{2/x} + d} = \frac{b}{d}$$

Hence $a = 2c$, $b = 2d$

$$\text{Now } bx^2 + (a - 2c)x - 2d = 0 \Rightarrow x^2 = 1.$$

5. Ans (A,B,D)

Since $g(x)$ is continuous for all $x \in \mathbb{R}$, $g(x)$ should be constant.

$$\text{Since } f(x) \in \left(2, \sqrt{26}\right) \Rightarrow a \geq \sqrt{26}, \text{ (as } \left[\frac{f(x)}{\sqrt{26}}\right]$$

$= 0$ for all $x \in \mathbb{R}$).

So least integral value of a is 6.

6. Ans (A,B)

$$\text{Given } y = \cos^{-1} \left(\frac{x^{2n} - 1}{x^{2n} + 1} \right)$$

Let $x^n = \tan \theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, then

$$y = \cos^{-1}(-\cos 2\theta) \Rightarrow y = \pi - \cos^{-1}(\cos 2\theta)$$

$$\Rightarrow y = \begin{cases} \pi + 2\theta, & -\frac{\pi}{2} < \theta < 0 \\ \pi - 2\theta, & 0 \leq \theta < \frac{\pi}{2} \end{cases}$$

$$\Rightarrow y = \begin{cases} \pi + 2\tan^{-1}x^n, & x^n < 0 \\ \pi - 2\tan^{-1}x^n, & x^n \geq 0 \end{cases}$$

$$\Rightarrow \frac{dy}{dx} = \begin{cases} \frac{2nx^{n-1}}{1+x^{2n}}, & x^n < 0 \\ \frac{-2nx^{n-1}}{1+x^{2n}}, & x^n \geq 0 \end{cases}$$

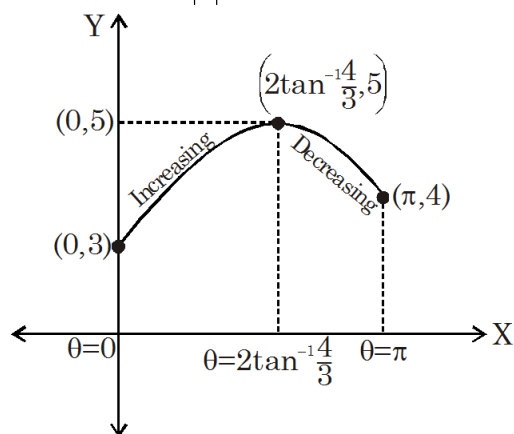
PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

(A) Let angle between $\vec{a}$ and $\vec{b}$ be θ

$$\text{we have } |\vec{a}| = |\vec{b}| = 1$$



Graph of $y = F(\theta)$ in $[0, \pi]$

$$\text{Now } |\vec{a} + \vec{b}| = 2 \cos \frac{\theta}{2} \text{ and}$$

$$|\vec{a} - \vec{b}| = 2 \sin \frac{\theta}{2}$$

$$\text{Consider } F(\theta) = \frac{3}{2} \left(2 \cos \frac{\theta}{2} \right) + 2 \left(2 \sin \frac{\theta}{2} \right)$$

$$\therefore F(\theta) = 3 \cos \frac{\theta}{2} + 4 \sin \frac{\theta}{2}, \quad \theta \in [0, \pi]$$

$$F'(\theta) = \frac{-3}{2} \sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2}$$

Now $F'(\theta) = 0$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{4}{3}$$

Clearly $F(0) = 3$

$$F\left(\theta = 2\tan^{-1}\frac{4}{3}\right) = 3\left(\frac{3}{5}\right) + 4\left(\frac{4}{5}\right) = \frac{9}{5} + \frac{16}{5} = \frac{25}{5} = 5$$

$$F(\theta = \pi) = 4$$

$$\therefore \text{Range} = [3, 5]$$

Hence possible integer(s) in the range of $F(\theta)$

in $[0, \pi]$ is 3 (viz., 3, 4 and 5)

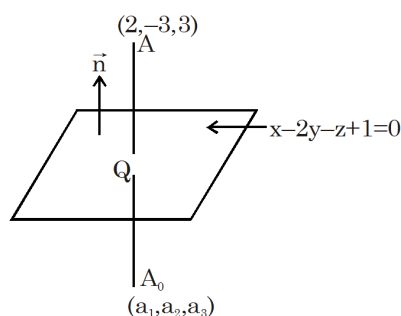
$$(B) \begin{bmatrix} \vec{V}_1 \vec{V}_2 \vec{V}_3 \end{bmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & 4 \\ 0 & 0 & 2\lambda - 1 \end{vmatrix} \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}$$

$$= 1(\lambda(2\lambda - 1)) \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = 0$$

$$\Rightarrow \lambda = 0, \frac{1}{2}$$

$$(\text{as } \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} \neq 0)$$

(C) We have $\vec{n} = \hat{i} - 2\hat{j} - \hat{k}$



Equation of line through A is

$$\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z-3}{-1} = \lambda$$

$$\Rightarrow x = \lambda + 2,$$

$$y = -(2\lambda + 3),$$

$$z = 3 - \lambda$$

$$\text{Hence } \lambda + 2 + 2(2\lambda + 3) - (3 - \lambda) + 1 = 0$$

$$\Rightarrow 6\lambda + 6 = 0 \Rightarrow \lambda = -1$$

So, Q is (1, -1, 4)

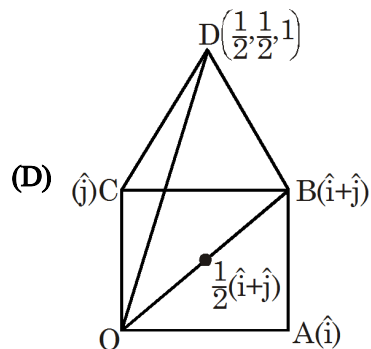
$$\text{Hence } a_1 + 2 = 2 \Rightarrow a_1 = 0$$

$$a_2 - 3 = -2 \Rightarrow a_2 = 1$$

$$a_3 + 3 = 8 \Rightarrow a_3 = 5$$

Image is (0, 1, 5)

Obviously distance of image from z-axis is 1



$$\vec{OD} = \vec{V}_1 = \frac{1}{2}(\hat{i} + \hat{j} + 2\hat{k})$$

$$\& \vec{OB} = \vec{V}_2 = (\hat{i} + \hat{j})$$

$$\therefore \cos \theta = \frac{\vec{V}_1 \cdot \vec{V}_2}{|\vec{V}_1| |\vec{V}_2|} = \frac{1+1}{\sqrt{2} \cdot \sqrt{6}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = \tan^{-1} \sqrt{2} \Rightarrow K = 2$$

8. Ans (C)

$$(P) f_1(x) = 1 + x \& f_2(x) = 1 - x$$

So, given expression is $(1+x)^{2010} + (1-x)^{2010}$

Hence, $N = 1006$

So, ${}^{(N-1000)}C_r = {}^6C_r$ is the greatest

when $r = 3$

$$(Q) t_{r+1} = {}^{18}C_r (9x)^r \left(\frac{-x^{-1/2}}{3} \right)^{18-r}$$

$$\text{So, } r + \frac{r-18}{2} = 0 \Rightarrow r = 6$$

$$\text{Thus, } t_7 = {}^{18}C_6 \times 9^6 \times \frac{1}{3^{12}} = {}^{18}C_6$$

$$\Rightarrow a = 1$$

$$(R) \sum_{x=2}^{\infty} \frac{1}{f_1(x)f_2(x)} = \sum_{x=2}^{\infty} \frac{1}{(1+x)(1-x)}$$

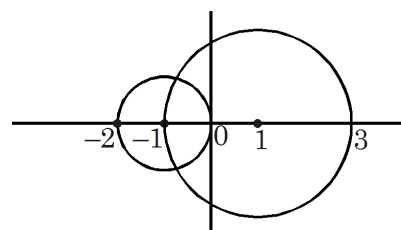
$$= - \sum_{x=2}^{\infty} \frac{1}{(x-1)(x+1)} = \frac{-1}{2} \sum_{x=2}^{\infty} \left(\frac{1}{x-1} - \frac{1}{x+1} \right)$$

$$= \frac{-1}{2} \left(1 + \frac{1}{2} \right) = \frac{-3}{4}$$

(S) Given pair of curves are

$$|Z+1| = 1 \text{ and } |Z-1| = 2.$$

They intersect at two distinct points



9. Ans (D)

$$(A) \cot^{-1}(\tan(-37^\circ)) = \cot^{-1}(\cot(90 + 37^\circ)) \\ = \cot^{-1}(\cot(127^\circ)) = 127^\circ$$

$$(B) \cos^{-1}(\cos(233^\circ)) = \cos^{-1}(\cos(360^\circ - 127^\circ)) \\ = \cos^{-1}(\cos(127^\circ)) = 127^\circ$$

$$(C) T = \sin \frac{\theta}{2}, \text{ where } \cos \theta = \frac{1}{9}; 1 - 2 \sin^2 \frac{\theta}{2} = \frac{1}{9}$$

$$\therefore 2 \sin^2 \frac{\theta}{2} = 1 - \frac{1}{9} = \frac{8}{9}$$

$$\Rightarrow \sin^2 \frac{\theta}{2} = \frac{4}{9}$$

$$\Rightarrow T = \sin \frac{\theta}{2} = \frac{2}{3}$$

$$(D) T = \cos \frac{\theta}{2}, \text{ where } \cos \theta = \frac{1}{8}$$

$$\text{i.e. } 2 \cos^2 \frac{\theta}{2} - 1 = \frac{1}{8}$$

$$\Rightarrow 2 \cos^2 \frac{\theta}{2} = 1 + \frac{1}{8} = \frac{9}{8}$$

$$\Rightarrow \cos^2 \frac{\theta}{2} = \frac{9}{16}$$

$$T = \cos \frac{\theta}{2} = \frac{3}{4}$$

10. Ans (A)

$$(P) \text{ Given } \cot^{-1}x + \cot^{-1}y + \cot^{-1}z = \frac{\pi}{4}$$

So $x, y, z > 1$

Hence,

$$\tan \left(\tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{y} + \tan^{-1} \frac{1}{z} \right) = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{\sum \frac{1}{x} - \frac{1}{xyz}}{1 - \sum \frac{1}{xy}} = 1$$

$$\Rightarrow \Sigma xy - 1 = xyz - \Sigma x$$

$$(Q) \sin^{-1}x = \frac{1}{2}, 2 \text{ (reject)}$$

$$\Rightarrow x = \frac{\pi}{6}$$

$$(R) \sin(\sin^{-1}2x - \cos^{-1}2x) = \sin(\sin^{-1}(6x - 2))$$

$$\Rightarrow (2x)^2 - (\sqrt{1 - 4x^2})^2 = 6x - 2$$

$$\Rightarrow 8x^2 - 6x + 1 = 0$$

$$\Rightarrow x = \frac{1}{2}, \frac{1}{4} \text{ (Both satisfies given equation)}$$

$$(S) \tan^{-1}(2 - \sqrt{3}) = \frac{\pi}{12}$$

$$\text{So, } 3 \times \frac{\pi}{12} - \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{1}{x}$$

$$\Rightarrow \tan^{-1} \frac{1}{x} = \tan^{-1} 1 - \tan^{-1} \frac{1}{3}$$

$$\Rightarrow \tan^{-1} \frac{1}{x} = \tan^{-1} \left(\frac{1 - \frac{1}{3}}{1 + \frac{1}{3}} \right) = \tan^{-1} \frac{1}{2}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (18.00)

$$\frac{3\lambda - 2\mu + 6}{2} = \frac{2\lambda - 4\mu + 8}{2} = \frac{\lambda - 3\mu + 4}{1}$$

On solving we get $\mu = 0$ and $\lambda = 2$

Hence $A(13, 11, 5); B(1, -1, -1)$

$$\Rightarrow AB = \sqrt{144 + 144 + 36} = \sqrt{324} = 18$$

2. Ans (1.00)

$$g'(x) = ae^{ax} + f'(x)$$

$$\Rightarrow g'(0) = a + f'(0) = a - 5 \quad \dots(1)$$

$$\text{Also, } g''(x) = a^2 e^{ax} + f''(x)$$

$$\Rightarrow g''(0) = a^2 + f''(0) = a^2 + 3 \quad \dots(2)$$

$$\text{Now, } g'(0) + g''(0) = 0$$

$$\Rightarrow a - 5 + a^2 + 3 = 0$$

$$\Rightarrow a^2 + a - 2 = 0$$

$$\Rightarrow (a + 2)(a - 1) = 0$$

$$\Rightarrow a = 1, -2$$

3. **Ans (3.00)**

$$f(x+y) = f(x)f(y) - g(x)g(y) \quad \dots(i)$$

put $x = y = 0$ in (i)

$$a = a^2 - b^2 \quad (f(0) = a, g(0) = b) \quad \dots(iii)$$

$$\text{Also, } g(x+y) = g(x)f(y) + f(x)g(y) \quad \dots(ii)$$

put $x = y = 0$ in (ii)

$$b = 2ab \quad \dots(iv)$$

from (iii) & (iv), we get only valid pair as $a = 1$,

$$b = 0$$

$$\text{i.e. } f(0) = 1 \text{ \& } g(0) = 0$$

$$\begin{aligned} \text{Now, } f'(x) &= \lim_{y \rightarrow 0} \frac{f(x+y) - f(x)}{y} \\ &= \lim_{y \rightarrow 0} f(x) \left(\frac{f(y) - f(0)}{y} \right) - g(x) \left(\frac{g(y) - g(0)}{y} \right) \end{aligned}$$

$$\Rightarrow f'(x) = f'(0)f(x) - g'(0)g(x)$$

$$\Rightarrow f'(x) = -g'(0)g(x) \quad (\text{as } f'(0) = 0) \quad \dots(v)$$

Similarly, from the second given relation

$$g'(x) = f'(0)g(x) + g'(0)f(x)$$

$$\Rightarrow g'(x) = g'(0)f(x) \quad (\text{as } f'(0) = 0) \quad \dots(vi)$$

Note $g'(0) \neq 0$, otherwise both $f(x)$ & $g(x)$

becomes constant

Now, (v) $\div$ (vi)

$$\frac{f'(x)}{g'(x)} = -\frac{g(x)}{f(x)} \Rightarrow f(x)f'(x) = -g(x)g'(x)$$

$$\Rightarrow 2f(x)f'(x) + 2g(x)g'(x) = 0$$

$$\text{Integrating :- } f^2(x) + g^2(x) = c = 1$$

$$(\text{as } f(0) = 1, g(0) = 0)$$

Now,

$$\begin{aligned} \frac{1 - (f^6 + g^6)}{f^2 g^2} &= \frac{1 - (f^2 + g^2)(f^4 + g^4 - f^2 g^2)}{f^2 g^2} \\ &= \frac{1 - (1 - 3f^2 g^2)}{f^2 g^2} = 3 \quad (\text{as } f^2 + g^2 = 1) \end{aligned}$$

4. **Ans (240.00)**

$$\text{Volume (V)} = \frac{1}{3} A_1 h_1 \Rightarrow h_1 = \frac{3V}{A_1}$$

$$\text{Similarly } h_2 = \frac{3V}{A_2}, h_3 = \frac{3V}{A_3} \text{ and } h_4 = \frac{3V}{A_4}$$

$$\begin{aligned} \text{So } (A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4) \\ = (A_1 + A_2 + A_3 + A_4) \left(\frac{3V}{A_1} + \frac{3V}{A_2} + \frac{3V}{A_3} + \frac{3V}{A_4} \right) \\ = 3V(A_1 + A_2 + A_3 + A_4) \left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4} \right) \end{aligned}$$

Now using A.M.-H.M inequality in $A_1, A_2, A_3,$

A_4 , we get

$$\begin{aligned} \frac{A_1 + A_2 + A_3 + A_4}{4} &\geq \frac{4}{\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4} \right)} \\ \Rightarrow (A_1 + A_2 + A_3 + A_4) \left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4} \right) &\geq 16 \end{aligned}$$

Hence the minimum value of

$$\begin{aligned} (A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4) \\ = 3V(16) = 48V = 48 \times 5 = 240 \end{aligned}$$

5. **Ans (0.75)**

We have $1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + \dots = 2$

$$\Rightarrow \frac{1}{1 - \sin(\cos^{-1} x)} = 2 \Rightarrow \frac{1}{2} = 1 - \sin$$

$$(\cos^{-1} x) \Rightarrow \sin(\cos^{-1} x) = \frac{1}{2}$$

$$\Rightarrow \cos^{-1} x = \frac{\pi}{6}$$

$$\Rightarrow x = \frac{\sqrt{3}}{2}$$

6. **Ans (3.00)**

$$\left(e^{x^2} g(x) \right)' = 2xe^{x^2} g'(x)$$

$$\Rightarrow (2x - 1)g'(x) = 2xg(x)$$

$$\Rightarrow \frac{g'(x)}{g(x)} = \frac{2x}{2x - 1}; x > \frac{1}{2}$$

Integrating, $g(x) = \sqrt{2x - 1} e^x$ (as $g(1) = e$)

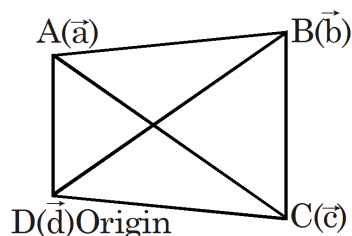
$$\text{So, } g(5) = 3e^5$$

7. Ans (4.00)

We have

$$\overrightarrow{AB}^2 + \overrightarrow{BC}^2 + \overrightarrow{CD}^2 + \overrightarrow{DA}^2 - \overrightarrow{AC}^2 - \overrightarrow{BD}^2 = \lambda \overrightarrow{PQ}^2$$

$$(\vec{b} - \vec{a})^2 + (\vec{c} - \vec{b})^2 + (\vec{d} - \vec{c})^2 + (\vec{a} - \vec{d})^2 - (\vec{c} - \vec{a})^2 - (\vec{d} - \vec{b})^2 = \lambda (\overrightarrow{PQ})^2$$



On simplifying gives

$$\text{LHS} = \vec{a}^2 + \vec{b}^2 + \vec{c}^2 - 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} + 2\vec{a} \cdot \vec{c},$$

$$\vec{d} = \vec{0}$$

$$= (\vec{a} + \vec{c} - \vec{b})^2$$

$$\text{again } \vec{P} = \frac{\vec{a} + \vec{c}}{2}; \quad \vec{Q} = \frac{\vec{b}}{2}$$

So,

$$\overrightarrow{PQ} = \frac{\vec{a} + \vec{c} - \vec{b}}{2} \Rightarrow 4(\overrightarrow{PQ})^2 = (\vec{a} + \vec{c} - \vec{b})^2$$

Hence, $\lambda = 4$

8. Ans (9.00)

Rearranging the given recurrence gives

$$\cot x_{n+1} - \cot x_n = \frac{1}{2^{n+1}}, \quad x \geq 1$$

Change n to k and sum the expression from k = 1

to k = n to yield a telescoping sum in the L.H.S.

and a geometric series in R.H.S., hence we are

led to

$$\cot x_{n+1} - \cot x_1 = \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{n+1}}$$

$$\therefore \cot x_{n+1} = 1 - \frac{1}{2^{n+1}} \text{ change 'n+1' to 'n' to}$$

$$\text{give } \cot x_n = 1 - \frac{1}{2^n}, \quad n \in \mathbb{N}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

PART TEST

21-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

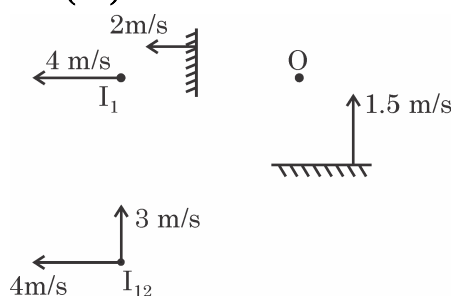
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	A	C	A	D	D	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	B	C	C	C	A	D	D	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	7.50	0.5	1.00	5.00	10.00	2.00	1.41	7.50	10.00	10.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	B	D	D	D	C	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	D	C	C	D	A	A	A	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	7.00	2.00	6.00	5.00	2.00	18.00	12.00	990.4	1.66 to 1.67	3.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	B	A	C	D	C	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	A	B	B	D	D	D	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3.00	1.00	3.00	5150.00	9.00	5.00	6.00	4.00	55.00	3.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I
1. Ans (C)

2. Ans (A)

 For normal vision $\frac{f_0}{f_e} = x$

 For near point vision $\frac{f_0}{D} \left(1 + \frac{D}{f_e}\right) = y$

$$\Rightarrow \frac{f_0}{D} + \frac{f_0}{f_e} = y$$

$$\frac{f_0}{D} = y - x$$

$$f_0 = (y - x)D$$

3. **Ans (A)**

Image of B after two reflection will coincide with B.

Now for A first at M_2 $u = -10\text{cm}$ $f = -20\text{cm}$

$$\frac{1}{V} + \frac{1}{-10} = \frac{1}{-20} \Rightarrow V = +20\text{cm}$$

Now at M_1 $u = -60\text{cm}$ $f = -20\text{cm}$

$$\frac{1}{V} - \frac{1}{60} = \frac{1}{-20} \Rightarrow V = -30\text{cm}$$

this again coincide with A.

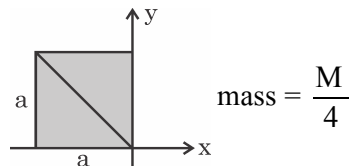
4. **Ans (C)**

Refer text.

5. **Ans (A)**

$$I_1 + I_2 = I_z = 7I_0$$

where I_0 is MI of one triangular plate about z-axis



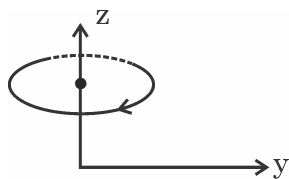
$$I_x = I_y = \frac{1}{3} \frac{M}{4} a^2$$

$$2I_0 = I_x + I_y = 2 \times \frac{1}{3} \frac{M}{4} a^2$$

$$\text{So } I = 7I_0 = \frac{7Ma^2}{12}$$

6. **Ans (D)**

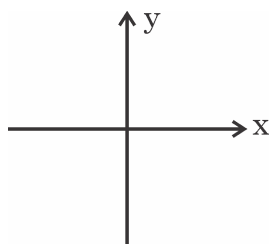
$$\vec{V} = 6 \cos 2t \hat{i} - 6 \sin 2t \hat{j}$$



$$\vec{L} = \vec{r} \times m\vec{V}$$

$$\vec{L} = (3 \sin 2t \hat{i} + 3 \cos 2t \hat{j}) \times 12 (\cos 2t \hat{i} - \sin 2t \hat{j})$$

$$= -36 \hat{k}$$



7. **Ans (D)**

Since glass slabs give constant shift so the velocity of image will be same as that of the object.

8. **Ans (B)**

$$\text{Angular acceleration of first rod} = \frac{mg\ell/2}{\frac{m\ell^2}{3}}$$

$$= \frac{3g}{2\ell}$$

For second rod

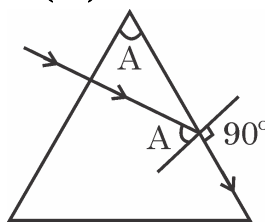
$$\alpha = \frac{mg\frac{\ell}{2} + mgx}{\frac{m\ell^2}{3} + mx^2} = \frac{3g}{2\ell}$$

$$\Rightarrow \ell_2 + 2\ell x = \ell_2 + 3x^2$$

$$3x^2 = 2\ell x$$

$$x = \frac{2\ell}{3}$$

9. **Ans (C)**



$$i = 0, e = 90^\circ$$

$$D = 30^\circ$$

$$i + e = A + D$$

$$A = 90^\circ - 30^\circ = 60^\circ$$

$$\mu = \frac{1}{\sin A} = \frac{2}{\sqrt{3}}$$

10. **Ans (B)**

$$E_1 = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2$$

$$= \frac{1}{2}mv^2 + \frac{1}{2} \frac{2}{5}mv^2$$

$$= \frac{7}{10}mv^2$$

$$E_2 = \frac{1}{2}m(2v)^2 + \frac{1}{2} \frac{2}{5}mv^2$$

$$= 2mv^2 + \frac{1}{5}mv^2$$

$$= \frac{11}{5}mv^2$$

$$\frac{E_1}{E_2} = \frac{7}{10} \times \frac{5}{11} = \frac{7}{22}$$

11. **Ans (C)**

General form of snell's law

$$\sin i = \mu \sin 90^\circ$$

$$\frac{1}{2} = \frac{1}{\sqrt{1-y^2}} \Rightarrow 1+y^2=4$$

$$y = \sqrt{3}$$

12. **Ans (B)**

$$\frac{1}{f_1} = (1.5 - 1) \left[\frac{1}{120} - \frac{1}{-120} \right]$$

$$= 0.5 \times \frac{2}{120} = \frac{1}{120}$$

$$f_1 = 120 \text{ cm} \quad f_m = -60 \text{ cm}$$

$$-\frac{1}{e_{eq}} = \frac{2}{f_1} - \frac{1}{f_m} \Rightarrow -\frac{1}{e_{eq}} = \frac{1}{60} - \frac{1}{-60} = \frac{1}{30}$$

$$f_{eq} = -30 \text{ cm}$$

13. **Ans (C)**

$$L_i = L_f$$

$$mu \frac{\ell}{4} = \left[\frac{m\ell^2}{12} + \frac{m\ell^2}{16} \right] \omega$$

$$mu \frac{\ell}{4} = m\ell^2 \frac{7}{48} \omega$$

$$\omega = \frac{12u}{7\ell}$$

15. **Ans (C)**

By conservation of angular momentum about point on ground

$$muR + 0 = m\omega R^2 + \frac{1}{2}mR^2\omega$$

$$muR = \frac{3}{2}mR^2\omega$$

$$\omega = \frac{2u}{3R}$$

16. **Ans (A)**

Case-I

$$u = -60 \text{ cm} \quad f = 20 \text{ cm} \quad v_1 = ?$$

$$\frac{1}{V_1} - \frac{1}{-60} = \frac{1}{20}$$

$$\frac{1}{V_1} = \frac{1}{20} - \frac{1}{60} \Rightarrow V_1 = 30 \text{ cm}$$

Case-II

$$u = -40 \text{ cm} \quad f = 20 \text{ cm} \quad v_2 = 40 \text{ cm}$$

$$d_{i/1} = d_i - d_1$$

$$10 = d_i - (-20)$$

$$d_i = -10 \text{ cm}$$

17. **Ans (D)**

Magnitude of torque and angular momentum are constant.

18. **Ans (D)**

Minimum distance between real object and its real image of $4f$.

$$D_{\min} = 4f$$

$$D \geq 4f$$

$$f \leq \frac{D}{4}$$

19. **Ans (D)**

$$\tau = I\alpha = \tau_0 e^{-\frac{kt}{2}}$$

$$\frac{d\omega}{dt} = \frac{\tau_0}{I} e^{-\frac{kt}{2}}$$

$$\omega = \frac{2\tau_0}{kI} \left[1 - e^{-\frac{kt}{2}} \right]$$

$$\text{Power} = \tau \cdot \omega$$

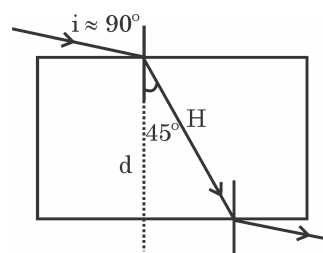
$$= \tau_0 e^{-\frac{kt}{2}} \frac{2\tau_0}{Ik} \left(1 - e^{-\frac{kt}{2}} \right)$$

For max power

$$e^{-\frac{kt}{2}} = 1 - e^{-\frac{kt}{2}}$$

$$t = \frac{2 \ln 2}{k} = \frac{\ln 4}{k}$$

20. **Ans (A)**



$$i_{\max} = 90^\circ$$

$$r = 45^\circ$$

$$H = \frac{d}{\cos 45} = d\sqrt{2}$$

$$c_m = \frac{C}{\sqrt{2}}$$

$$\text{Time} \frac{H}{c_m} = \frac{d\sqrt{2}}{c/\sqrt{2}} = \frac{2d}{c}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (7.50)**

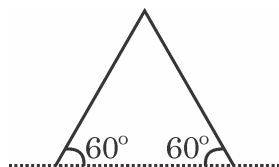
$$L_i = \left(\frac{1}{2}MR^2 + mR^2 \right) \omega$$

$$L_f = \left(\frac{1}{2}MR^2 + md^2 \right) \omega' \quad d = 3m$$

$$\left[\frac{1}{2}2(.5)^2 + 1(.5)^2 \right] \omega = \left[\frac{1}{2}2(5)^2 + 1(3)^2 \right] \omega'$$

$$\frac{50\omega}{34} = \omega'$$

2. Ans (0.5)



$$I = 2 \frac{1}{3} M \ell^2 \sin^2 60^\circ$$

$$= \frac{2}{3} M \ell^2 \cdot \frac{3}{4}$$

$$= \frac{M \ell^2}{2} = 0.5 M \ell^2$$

3. Ans (1.00)

$$V_{AB} = \omega r_{AB}$$

$$8 = \omega \cdot 8$$

$$\omega = 1 \text{ rad/s}$$

4. Ans (5.00)

$$M = 1 + \frac{D}{f} = 6$$

$$\frac{D}{f} = 5$$

$$f = \frac{D}{5} = 5 \text{ cm}$$

5. Ans (10.00)

$$L_f - L_i = \int \tau dt$$

$$I\omega - 0 = \frac{1}{2} 2 \times 4$$

$$\omega = \frac{4}{0.4} = 10 \text{ rad/s}$$

6. Ans (2.00)

$$D = (\mu - 1) A$$

$$= (1.5 - 1) 4^\circ$$

$$= 2^\circ$$

7. Ans (1.41)

The ray will suffer refraction at $i = 45^\circ$ and TIR

at $i = 67.5^\circ$ so For $i = 45^\circ D_1 = 45 - r$

For $i = 67.5^\circ$

$$D_2 = 180 - 2 \times 67.5$$

$$= 180 - 135$$

$$= 45^\circ$$

$$r - 45^\circ = 45^\circ$$

$$r = 90^\circ$$

From snell's law

$$\frac{1}{\mu} = \frac{\sin i}{\sin r} = \frac{\sin 45}{\sin 90} = \frac{1}{\sqrt{2}}$$

$$\mu = \sqrt{2} = 1.414$$

8. Ans (7.50)

It will ascend through maximum height if surface is sufficiently rough

$$\frac{1}{2} mv^2 \left(1 + \frac{k^2}{R^2} \right) = mgh$$

$$\frac{1}{2} mv^2 \frac{3}{2} = mgh$$

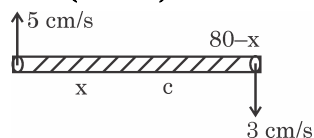
$$h = \frac{3v^2}{4g} = 7.5 \text{ m}$$

9. Ans (10.00)

$$m = \frac{f - V}{f}$$

$$m = 0 \text{ if } v = f = 10 \text{ cm}$$

10. Ans (10.00)



$$W = \frac{5}{x} = \frac{3}{80-x}$$

$$\Rightarrow 400 = 3x + 5x$$

$$x = \frac{400}{8} = 50 \text{ cm}$$

$$\text{Distance from centre} = 50 - 40 = 10 \text{ cm}$$

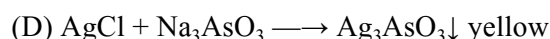
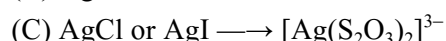
PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)



(B) AgI is more covalent so less soluble



2. Ans (D)

AgCl, PbCl₂, Hg₂Cl₂ are precipitate CuCl₂ is soluble.

3. Ans (C)

AgI ↓ Yellow

Hg₂I₂ ↓ Green

PbI₂ ↓ Yellow

4. Ans (B)

In presence of excess of CO_3^{2-} , Mg^{2+} will also be precipitated as MgCO_3

5. **Ans (D)**

(A) $\text{Mg}(\text{NH}_4)\text{PO}_4$ dissolves due to protonation of PO_4^{3-}

(B) $\text{Mg}(\text{OH})_2 + \text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O}$

(C) NH_4Cl & $(\text{NH}_4)_2\text{CO}_3$ are Vth group reagents

(D) $\text{Mg}(\text{OH})_2 \xrightarrow{\text{titan yellow}} \text{Red}$

6. **Ans (D)**

(D) $\text{PO}_4^{3-} + \text{MgCl}_2 + \text{NH}_4\text{Cl} + \text{NH}_3 \rightarrow \text{Mg}(\text{NH}_4)\text{PO}_4 \downarrow \text{white}$

7. **Ans (D)**

(i) $\text{NO}_2^- + \text{I}^- + \text{CH}_3\text{COOH} \longrightarrow \text{I}_2 + \text{NO} + \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$

(ii) $\text{NO}_2^- + \text{NH}_4^+ \xrightarrow{\Delta} \text{N}_2 + 2\text{H}_2\text{O}$

(iii) $\text{NO}_2^- + \text{NH}_2\text{CONH}_2 \xrightarrow{\text{H}^+} \text{N}_2 + \text{CO}_2 + \text{H}_2\text{O}$

(iv) $\text{NO}_2^- + \text{NH}_2\text{SO}_3\text{H} \longrightarrow \text{N}_2 + \text{SO}_4^{2-} + \text{H}_2\text{O}$

8. **Ans (C)**

Initially $P_{\text{N}_2} = P_{\text{H}_2}$

$$\frac{nRT}{10} = \frac{10RT}{100}$$

Finally

$P_{\text{N}_2} = P_{\text{H}_2}$

$$\frac{1RT}{10-x} = \frac{11RT}{100+x}$$

$$100+x = 110-11x$$

$$12x = 10$$

$$x = \frac{5}{6} = 0.83$$

9. **Ans (B)**

$$Z = 1 - \frac{a}{V_m RT}$$

$$P V_m = RT - \frac{a}{V_m}$$

$$RT = 0.0821 \times 300$$

$$= 24.63$$

10. **Ans (D)**

$$\text{SC} : r = \frac{a}{2} = 2\text{\AA}$$

$$\text{BCC} : r = \frac{\sqrt{3}a}{4} = \sqrt{3}\text{\AA}$$

$$\text{FCC} : r = \frac{\sqrt{2}a}{4} = \sqrt{2}\text{\AA}$$

11. **Ans (A)**

Finally $P_{\text{H}_2\text{O}(v)} = 0.4 - 0.25 = 0.15 \text{ atm}$

$$0.15 = \frac{n \frac{1}{12} \cdot 300}{4}$$

$$n = \frac{0.6}{25}$$

$$\Rightarrow \text{mass of H}_2\text{O} \ell = \frac{0.6}{25} \times 18\text{g}$$

12. **Ans (D)**

Higher the inter molecular force of attraction, higher will be T_c .

13. **Ans (C)**

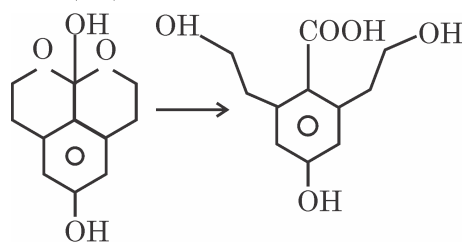
$$\lambda \propto \frac{T}{P d^2}$$

14. **Ans (C)**

$\text{LiCl} \rightarrow \text{Pink}$

Rest options are correct

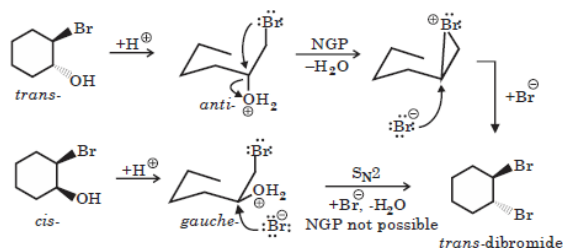
15. **Ans (D)**



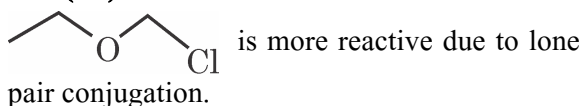
16. **Ans (A)**

Oxidative cleavage of double bond then thermal decarboxylation of beta carbonyl carboxylic acid.

17. Ans (A)



19. Ans (B)



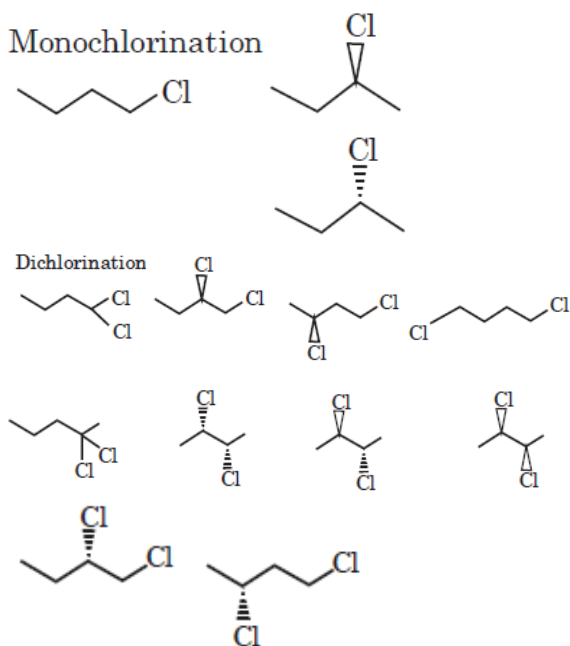
20. Ans (B)

Sulphur is strongest nucleophile due to size. Strong base are good nucleophile (if charge is present on same type of atom)

PART-2 : CHEMISTRY

SECTION-II

1. Ans (7.00)



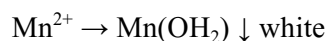
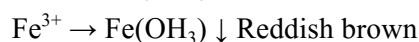
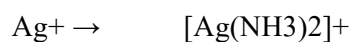
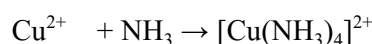
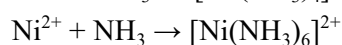
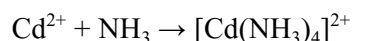
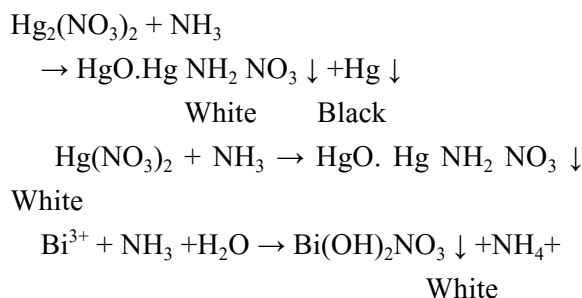
2. Ans (2.00)

- (i) NH_2 is the poorest leaving group.
(iii) Rate of reaction is also tripled.

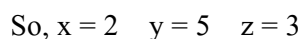
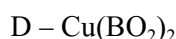
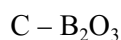
4. Ans (5.00)

(1), (2), (3), (7), (8) give primary alcohol as a product.

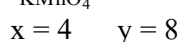
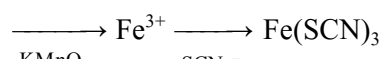
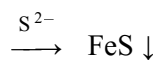
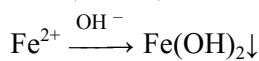
5. Ans (2.00)



6. Ans (18.00)



7. Ans (12.00)



8. Ans (990.4)

$V = V_{\text{container}} - nb$

$$b = 4 \times \frac{4}{3} \pi \left[\left(\frac{3}{\pi} \right)^{\frac{1}{3}} 10^{-8} \right]^3 6 \times 10^{23}$$

$$= 9.6 \text{ ml mol}^{-1}$$

9. **Ans (1.66 to 1.67)**

$$\begin{aligned} \text{mps} &= \sqrt{\frac{2p}{d}} = \sqrt{\frac{2 \times 0.2 \times 10^5}{1.44}} \\ &= \frac{2}{1.2} \times 10^2 \text{ m/sec} \\ &= 1.67 \times 10^2 \text{ m/sec} \end{aligned}$$

10. **Ans (3.00)**

$h = 4\sqrt{\frac{2}{3}}R$ so this plane is at $h/4$ height from layer A which will pass through center of 3 octahedral voids.

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

Vector parallel to 1st given line

$$\vec{a} = 3\hat{i} - 16\hat{j} + 7\hat{k}$$

Vector parallel to 2nd given line

$$\vec{b} = 3\hat{i} + 8\hat{j} - 5\hat{k}$$

$$\begin{aligned} \vec{c} &= \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} \\ &= \hat{i} (80 - 56) - \hat{j} (15 + 21) + \hat{k} (24 + 48) \\ &= 24\hat{i} + 36\hat{j} + 72\hat{k} \end{aligned}$$

$$\therefore \text{Required line } \frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}.$$

2. **Ans (B)**

for existence of $\sin^{-1}x$

$$0 \leq x^2 + 3x + 3 \leq 1$$

and for existence of $\sec^{-1}x$

$$x^2 + 3x + 3 \geq 1$$

$$\therefore x^2 + 3x + 3 = 1 \Rightarrow x^2 + 3x + 2 = 0 \Rightarrow x = -1, -2$$

$$\text{but } x \neq -1, \therefore x = -2$$

$$\therefore 2 \cos^{-1}\left(\frac{x}{2}\right) + \operatorname{cosec}^{-1}\left(\frac{x}{2}\right) = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$$

3. **Ans (D)**

$$\begin{vmatrix} 1 & -1 & -1 \\ 2 & 3 & \lambda \\ 3 & 2 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

4. **Ans (B)**

$$\text{As, } \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2} \quad \forall x \in \mathbb{R}$$

$$(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = 7 \left(\frac{\pi^3}{8}\right)$$

$$\Rightarrow (\sin^{-1}x + \cos^{-1}x)^3 - 3(\sin^{-1}x)(\cos^{-1}x)$$

$$(\sin^{-1}x + \cos^{-1}x) = 7 \left(\frac{\pi}{2}\right)^3$$

$$\Rightarrow \left(\frac{\pi}{2}\right)^3 - 3(\sin^{-1}x) \frac{\pi}{2}(\cos^{-1}x) = 7 \left(\frac{\pi}{2}\right)^3$$

$$\Rightarrow (\sin^{-1}x)(\cos^{-1}x) = \frac{-\pi^2}{2} \dots\dots\dots(1)$$

Now, as maximum value of $\cos^{-1}x$ is

π ($\cos^{-1}x \geq 0$) and minimum value of $\sin^{-1}x$

is $\frac{-\pi}{2}$ and this happens at the same x i.e., at

$$x = -1.$$

Hence, $x = -1$ is the only solution.

5. **Ans (A)**

$$(\vec{r} - \vec{q}) \times \vec{p} = 0 \Rightarrow \vec{r} - \vec{q} = \lambda \vec{p} \dots\dots\dots(1)$$

$$\text{now } \vec{r} \cdot \vec{s} = 1$$

$$\Rightarrow (\lambda \vec{p} + \vec{q}) \cdot \vec{s} = 1 \quad \lambda = - \left(\frac{\vec{q} \cdot \vec{s} - 1}{\vec{p} \cdot \vec{s}} \right) \dots\dots\dots(2)$$

Put λ from (2) in (1) to get the result

6. **Ans (C)**

$$f(x) = \begin{cases} 2x^2 + 3, & -2 < x < 2 \\ x + 4, & 2 < x < 4 \end{cases},$$

$$\therefore \lim_{x \rightarrow 2^-} (2x^2 + 3) = 11 \text{ and } \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x + 4) = 6$$

$$(x + 4) = 6$$

Hence, the required equation will be

$$x^2 - (\text{sum of roots})x + (\text{Product of roots}) = 0$$

$$\text{i.e. } x^2 - 17x + 66 = 0$$

7. **Ans (D)**

$$T_1 = \tan^{-1}3 - \tan^{-1}1$$

$$T_2 = \tan^{-1}4 - \tan^{-1}2$$

$$T_3 = \tan^{-1}5 - \tan^{-1}3$$

$$\vdots$$

$$T_n - 1 = \tan^{-1}(n+1) - \tan^{-1}(n-1)$$

$$T_n = \tan^{-1}(n+2) - \tan^{-1}(n)$$

$$\therefore S_\infty = \pi - (\tan^{-1}(1) + \tan^{-1}(2)) = \tan^{-1}(3)$$

8. **Ans (C)**

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 0 & 3 & 2 \end{vmatrix} = -3\hat{i} - 4\hat{j} + 6\hat{k}$$

$$\text{Plane : } 3(x-1) + 4(y-2) - 6(z+1) = 0$$

$$3x + 4y - 6z - 10 = 0$$

9. **Ans (B)**

$$\text{Given } \frac{c+1}{c-1} = \frac{(x^2+x+1)^2}{x^4+x^2+1} =$$

$$\frac{(x^2+x+1)^2}{(x^2+x+1)(x^2-x+1)} = \frac{x^2+x+1}{x^2-x+1}$$

$$\Rightarrow c = \frac{x^2+1}{x} = x + \frac{1}{x} = f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right)$$

$$\therefore \left[f(f(x)) = \frac{1-f(x)}{1+f(x)} = \frac{1-\frac{1-x}{1+x}}{1+\frac{1-x}{1+x}} = x \right]$$

10. **Ans (D)**

$$|\vec{a} + \vec{b}|^2 = |\vec{a} - 2\vec{b}|^2$$

$$|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{a}|^2 + 4|\vec{b}|^2 - 4\vec{a} \cdot \vec{b}$$

$$6\vec{a} \cdot \vec{b} = 3|\vec{b}|^2$$

$$\vec{a} \cdot \vec{b} = \frac{1}{2}|\vec{b}|^2$$

$$\frac{1}{2}|\vec{b}|^2 + \frac{4}{|\vec{b}|^2 + 2}$$

$$\frac{1}{2}|\vec{b}|^2 + \frac{2}{2} + \frac{4}{|\vec{b}|^2 + 2}$$

Apply A.M. ≥ G.M.

$$\text{Min value} = 2\sqrt{2} - 1$$

11. **Ans (A)**

$$\lim_{x \rightarrow 0} [\cos 3x + a^3 \sin(b^6 x)]^{\frac{8}{x}}$$

$$= e^{\lim_{x \rightarrow 0} [\cos 3x + a^3 \sin(b^6 x) - 1] \frac{8}{x}}$$

$$= 8e^{\lim_{x \rightarrow 0} \left[-\left(\frac{1 - \cos 3x}{x}\right) + \frac{a^3 \sin(b^6 x)}{x} \right]}$$

$$= 8e^{\lim_{x \rightarrow 0} \left[-\frac{2\sin^2\left(\frac{3x}{2}\right)}{x} + \frac{a^3 \sin(b^6 x)}{x} \right]}$$

$$= e^{0+8a^3b^6}$$

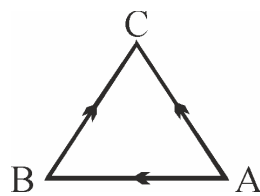
$$= e^{512} \text{ (given)}$$

$$\therefore a^3 b^6 = 64$$

$$\Rightarrow (ab^2)^3 = (4)^3$$

$$= ab^2 = 4$$

12. **Ans (A)**



Position C vector w.r.t

$$\vec{A} = \vec{AC} = \vec{AB} + \vec{BC} = \hat{i} + \hat{j} + \hat{i} - \hat{j} = 2\hat{i}.$$

13. **Ans (A)**

$$\text{Let } t = \sin^{-1}x \text{ as } x \in \left(0, \frac{1}{2}\right]$$

$$\text{Hence } t \in \left(0, \frac{\pi}{6}\right]$$

$$\therefore \text{equation is } (t)^{\frac{\pi}{2}-t} = \left(\frac{\pi}{2}-t\right)^t$$

$$\Rightarrow \left(\frac{\pi}{2}-t\right) \ln t = t \ln \left(\frac{\pi}{2}-t\right)$$

$$\therefore \frac{\ln t}{t} = \frac{\ln \left(\frac{\pi}{2}-t\right)}{\left(\frac{\pi}{2}-t\right)}$$

Since the $f(x) = \frac{\ln x}{x}$ is increasing and one

one function for $x \in (0, e]$

$$\text{Hence } t = \frac{\pi}{2} - t$$

$$\therefore t = \frac{\pi}{4} \text{ i.e. No solution } \Rightarrow \text{A.}$$

14. **Ans (B)**

$$\vec{r} = \lambda_1 \vec{r}_1 + \lambda_2 \vec{r}_2 + \lambda_3 \vec{r}_3$$

$$\Rightarrow 2\vec{a} - 3\vec{b} + 4\vec{c} = \lambda_1 (\vec{a} - \vec{b} + \vec{c}) + \lambda_2$$

$$(\vec{b} + \vec{c} - \vec{a}) + \lambda_3 (\vec{c} + \vec{a} + \vec{b})$$

$$\therefore \lambda_1 - \lambda_2 + \lambda_3 = 2$$

$$\Rightarrow -\lambda_1 + \lambda_2 + \lambda_3 = -3$$

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 = 4$$

$$\Rightarrow \lambda_1 = 7/2, \lambda_2 = 1 \text{ and } \lambda_3 = -1/2$$

15. Ans (B)

The domain of $\sqrt[3]{\sin x}$ is $\mathbb{R}$.

The domain of $\frac{1}{\sqrt{\sin x}} = [x : \sin x > 0]$

$$= \{(2k\pi, (2k+1)\pi) : k \in \mathbb{I}\}$$

The domain of $\log_{10} \frac{x-5}{x^2-10x+24}$

$$= \log_{10} \frac{x-5}{(x-6)(x-4)}$$

$$= \{x : x \neq 5, x > 5, x > 6, x > 4 \text{ or}$$

$$x \neq 5, x > 5, 4 < x < 6\}$$

$$= (4, 5) \cup (6, \infty)$$

Thus the domain of $f(x)$

$$= \{(2k\pi, (2k+1)\pi) : k \in \mathbb{I}\} \cap \{(4, 5) \cup (6, \infty)\}$$

$$= \{(2k\pi, (2k+1)\pi) : k \in \mathbb{N}\}$$

(since $(4, 5)$ does not intersect $\{(2k\pi, (2k+1)\pi) : k \in \mathbb{I}\}$)

16. Ans (D)

$$\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 3$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^3(1-a) - bx^2 - ax + (1-b)}{x^2 + 1} = 3$$

$$\Rightarrow 1-a=0 \text{ and } -b=3$$

$$\Rightarrow a=1, b=-3$$

17. Ans (D)

$$\text{Let } f(x) = \frac{1 - \cos 2x \sqrt{\cos 3x}}{8x^2}$$

$$= \frac{1 - \cos 2x \sqrt{\cos 3x}}{8x^2} \times \frac{1 + \cos 2x \sqrt{\cos 3x}}{1 + \cos 2x \sqrt{\cos 3x}}$$

$$= \frac{1 - \cos^2 2x \cos 3x}{8x^2} \times \frac{1}{1 + \cos 2x \sqrt{\cos 3x}}$$

$$\therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{(1 - \cos^2 2x (1 - 2\sin^2 3x/2))}{8x^2}$$

$$\times \lim_{x \rightarrow 0} \frac{1}{1 + \cos 2x \sqrt{\cos 3x}}$$

$$= \lim_{x \rightarrow 0} \left(\frac{1 - \cos^2 2x + 2\cos^2 2x \sin^2 3x/2}{8x^2} \right) \times \frac{1}{2}$$

$$= 1/4 + \left(\frac{2\cos^2 2x \sin^2 3x/2}{8x^2} \right) \times \frac{1}{2}$$

$$\frac{17}{32} = \frac{k_1}{k_2}$$

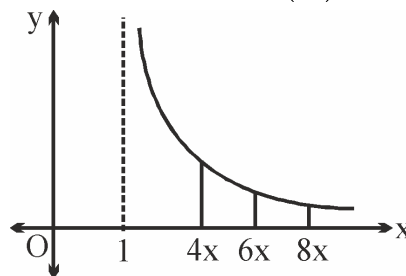
18. Ans (D)

As $f(x)$ is a continuous decreasing positive function on $(1, \infty)$, so

$$0 < f(8x) < f(5x) < f(x)$$

$$\Rightarrow 0 < 1 < \frac{f(5x)}{f(8x)} < \frac{f(x)}{f(8x)}$$

$$\Rightarrow \lim_{x \rightarrow \infty} 1 \leq \lim_{x \rightarrow \infty} \frac{f(5x)}{f(8x)} \leq \lim_{x \rightarrow \infty} \frac{f(x)}{f(8x)}$$



$\therefore$ By squeeze play theorem,

$$\lim_{x \rightarrow \infty} \frac{f(5x)}{f(8x)} = 1$$

19. Ans (B)

$$\frac{x^2 - 1}{x^2 + 1} = y$$

$$\therefore x^2 = \frac{1+y}{1-y}$$

Range is $[-1, 1)$ and it is many one

20. Ans (A)

$$\lim_{x \rightarrow 1} \frac{a\sqrt{x+3} - b}{x-1} = \frac{1}{4}$$

$$2a - b = 0 \Rightarrow b = 2a \quad \dots(1)$$

$$\lim_{x \rightarrow 1} \frac{a(\sqrt{x+3} - 2)}{x-1} = \frac{1}{4}$$

$$\lim_{x \rightarrow 1} \frac{a}{(\sqrt{x+3} + 2)} = \frac{1}{4}$$

$$a = 1, b = 2$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3.00)**

$$\lim_{x \rightarrow 0} \frac{(5^x - 1)^3}{\sin\left(\frac{x}{a}\right) \log\left(1 + \frac{x^2}{3}\right)}$$

$$= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x}\right)^3 \cdot \frac{x^3}{\left(\frac{x}{a}\right) \frac{\sin\left(\frac{x}{a}\right)}{\left(\frac{x}{a}\right)}} \times \frac{1}{\frac{\log\left(1 + \frac{x^2}{3}\right)}{\frac{x^2}{3}} \times \frac{x^2}{3}}$$

$$= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x}\right)^3 \cdot \frac{1 \times a}{\left(\frac{\sin\left(\frac{x}{a}\right)}{\frac{x}{a}}\right)} \cdot \frac{3}{\frac{\log\left(1 + \frac{x^2}{3}\right)}{\frac{x^2}{3}}}$$

$$\left(\text{using } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1, \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1\right)$$

$$\text{and } \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

$$= 3a(\log_e 5)^3$$

$$\text{Now, } f(0) = 9(\log_e 5)^3$$

Since, $f(x)$ is continuous at $x = 0$

$$\therefore 9(\log_e 5)^3 - 3a(\log_e 5)^3 \Rightarrow a = 3$$

2. **Ans (1.00)**

$$f(x) = f^{-1}(x)$$

$$f(x) = x$$

$$(x+1)^2 - 7 = x$$

$$x^2 + x - 6 = 0$$

$$x = 2, -3$$

3. **Ans (3.00)**

$$T = \frac{6\pi^2}{\sqrt{2}} = \lambda\pi^2; \frac{\lambda}{\sqrt{2}} = 3$$

4. **Ans (5150.00)**

$$f(x) - f(y) \geq \ln x - \ln y + x - y \quad \dots(1)$$

$$x \rightarrow y \text{ and } y \rightarrow x,$$

$$f(y) - f(x) \geq \ln y - \ln x + y - x$$

$$\text{i.e. } f(x) - f(y) \leq \ln x - \ln y + x - y \quad \dots(2)$$

from (1) and (2)

$$f(x) - f(y) = \ln x - \ln y + x - y \quad \dots(3)$$

$$\therefore f(x) - f(y) = (\ln x + x) - (\ln y + y)$$

$$\text{hence } f(x) = \ln x + x$$

$$\text{Alternatively : } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$(x \rightarrow x+h, y \rightarrow x \text{ in (3)})$$

$$= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x + h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(\frac{x+h}{x}\right) + h}{h}$$

$$= \lim_{h \rightarrow 0} \left[1 + \frac{\ln\left(1 + \left(\frac{h}{x}\right)\right)}{x\left(\frac{h}{x}\right)} \right]$$

$$= 1 + \frac{1}{x} \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t}$$

$$f'(x) = 1 + \frac{1}{x}$$

$$\therefore g(x) = 1 + \frac{1}{x}$$

$$\sum_{n=1}^{100} g\left(\frac{1}{n}\right) = \sum_{n=1}^{100} (1+n)$$

$$= 100 + (1 + 2 + 3 + 4 + \dots + 100)$$

$$= 100 + 5050 = 5150$$

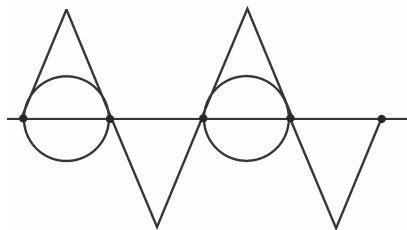
5. **Ans (9.00)**

$$\vec{a} \times \vec{b} = \hat{i} - 4\hat{j} + 2\hat{k}$$

$$[\vec{a} \vec{b} \vec{c}] = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\text{Maximum value} = |\vec{a} \times \vec{b}| |\vec{c}| = 2\sqrt{21}$$

6. Ans (5.00)



7. Ans (6.00)

$$\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{b}) \Rightarrow \vec{a} \times (\vec{b} - 2\vec{c}) = 0 \Rightarrow \vec{b} - 2\vec{c} = \alpha \vec{a}$$

$$\text{squaring } b^2 + 4c^2 - 4\vec{b} \cdot \vec{c} = \alpha^2 a^2$$

$$\Rightarrow 16 + 4 - 4 \cdot 4 \cdot 1 \cdot \frac{1}{4} = \alpha^2$$

$$\Rightarrow \alpha = \pm 4 \Rightarrow \vec{b} = 2\vec{c} \pm 4\vec{a} \Rightarrow |\ell| + |\mu| = 6$$

8. Ans (4.00)

$$f'(4) = \lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h} =$$

$$\lim_{h \rightarrow 0} \frac{f(4+h) - f(4+0)}{h} =$$

$$\lim_{h \rightarrow 0} \frac{2f(4)f(4) - 2f(4)f(0)}{h} =$$

$$(\text{using } f(x+y) = 2f(x)f(y))$$

$$= 2f(4) \cdot \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = 2f(4) \cdot f'(0) =$$

$$2 \times 2 \times 1 = 4$$

9. Ans (55.00)

The given equation is

$$f(x-1) + f(x+1) = \sqrt{2}f(x) \quad \dots (i)$$

Replace x by x - 1 in (i)

$$f(x-2) + f(x) = \sqrt{2}f(x-1) \quad \dots (ii)$$

Replace x by x + 1 in (ii)

$$f(x) + f(x+2) = \sqrt{2}f(x+1) \quad \dots (iii)$$

Adding (ii) and (iii), we have

$$f(x-2) + f(x+2) + 2f(x) = \sqrt{2}(f(x-1) + f(x+1))$$

$$= \sqrt{2} \sqrt{2} f(x) = 2f(x)$$

$$\Rightarrow f(x-2) = -f(x+2)$$

Replacing x by x + 2, we have

$$f(x) = -f(x+4) = -(f(x+8)) = f(x+8)$$

$$\sum_{r=0}^{10} f(3+8r) = 11 \times f(3) = 11 \times 5 = 55$$

10. Ans (3.00)

$$x_1 = \lambda + 1$$

$$y_1 = 8\lambda + 2$$

$$z_1 = 9\lambda + 3$$

$$\lambda + 1 + 2(8\lambda + 2) - 9\lambda - 3 = 0$$

$$\lambda + 1 + 16\lambda + 4 - 9\lambda - 3 = 0$$

$$8\lambda = -2 \Rightarrow \lambda = -\frac{1}{4}$$

$$\frac{\alpha + 1}{2} = \frac{3}{4}$$

$$\Rightarrow \alpha = \frac{3}{2} - 1 = \frac{1}{2}$$

$$\frac{\beta + 2}{2} = 0$$

$$\Rightarrow \beta = -2$$

$$\frac{\gamma + 3}{2} = -\frac{9}{4} + 3 = \frac{3}{4}$$

$$\Rightarrow \gamma + 3 = \frac{3}{2}$$

$$\Rightarrow \gamma = -\frac{3}{2}$$

$$\alpha + \beta + \gamma = \frac{1}{2} - 2 - \frac{3}{2}$$

$$= \frac{1 - 4 - 3}{2}$$

$$= -\frac{6}{2} = -3$$